

Viral Hepatitis — Vaccines & Post-Exposure Prophylaxis



1. Active vaccines shelf

WHAT YOU SEE

Left 'ACTIVE — VACCINES' shelf banner: slow start, long protection, memory

WHAT IT ENCODES

Active immunization (vaccine) presents inactivated antigen so the host mounts its own antibody and memory response: SLOW onset (weeks, requires multiple doses) but DURABLE, long-lasting protection with immunologic memory. Both HAV and HBV vaccines are inactivated/recombinant (not live), so they are safe in pregnancy and immunocompromise.

BOARD TRAP

Vaccine = slow but long-lasting; it is for pre-exposure protection and longer-term immunity, but its onset is too slow to be sole protection right after a high-risk exposure (then add passive Ig).

2. HAV vaccine (Havrix/Vaqta)

WHAT YOU SEE

HAV shelf with HAVRIX / VAQTA vials, 2-dose

WHAT IT ENCODES

HAV vaccine (Havrix, Vaqta) is an inactivated 2-dose series (0 and 6-12 months). It is the PREFERRED post-exposure prophylaxis for healthy people aged 12 months-40 years after HAV exposure, given within 2 weeks, and is routine for travelers and chronic liver disease.

BOARD TRAP

For a healthy exposed adult ≤40, HAV VACCINE (not immunoglobulin) is the preferred PEP — choosing Ig for a healthy young exposed adult is the trap. Ig is reserved for the very young, >40, immunocompromised, or chronic liver disease.

3. HBV vaccine (Engerix/Recombivax)

WHAT YOU SEE

HBV shelf with ENGERIX-B / RECOMBIVAX vials, 3-dose

WHAT IT ENCODES

HBV vaccine (Engerix-B, Recombivax HB) is recombinant HBsAg given as a 3-dose series (0, 1, 6 months). It produces protective anti-HBs ONLY (titer ≥10 mIU/mL) and never anti-HBc, since it contains no core antigen. Universally recommended for all infants and unvaccinated adults.

BOARD TRAP

Vaccine yields ISOLATED anti-HBs — anti-HBc stays negative. So a person with anti-HBs+ but anti-HBc-negative is vaccinated, not recovered from natural infection. Non-responders (anti-HBs <10) get a repeat series.

4. Heplisav-B 2-dose

WHAT YOU SEE

Glowing HEPLISAV-B vial center-left, '2 doses' star

WHAT IT ENCODES

Heplisav-B is a CpG-adjuvanted recombinant HBV vaccine requiring only 2 doses one month apart (vs 3 for Engerix/Recombivax), with higher seroprotection rates — a useful option for adults, including some prior non-responders.

BOARD TRAP

Don't assume every HBV vaccine is 3 doses — Heplisav-B is a 2-dose adult schedule. Matching the right number of doses to the product is the detail being tested.

5. HCV — no vaccine

WHAT YOU SEE

Top placard 'HCV: NO VACCINE'

WHAT IT ENCODES

There is NO vaccine and NO effective immunoglobulin for HCV (its high genetic variability has thwarted vaccine development). Prevention relies on harm reduction and universal screening; post-exposure management is RNA monitoring and early DAA treatment if infection is confirmed.

BOARD TRAP

Classic trap: 'give HCV immunoglobulin (or vaccine) after a needlestick' is WRONG — neither exists. The correct post-needlestick HCV approach is baseline and follow-up HCV RNA/antibody testing, then treat if it converts.

6. HDV covered by HBV vaccine

WHAT YOU SEE

Top-right 'HDV: use HBV vaccine — protects both'

WHAT IT ENCODES

There is no separate HDV vaccine, but HBV vaccination also prevents HDV, because HDV is a defective virus that needs HBsAg to infect — no HBV means no HDV.

BOARD TRAP

Don't look for a dedicated 'delta vaccine' — preventing HBV (via HBV vaccine) prevents HDV. However, the HBV vaccine cannot protect an already chronic HBsAg+ carrier from HDV SUPERinfection (they're already HBsAg+).

7. HBIG + vaccine = PEP

WHAT YOU SEE

Center 'HBIG + VACCINE' bench, immediate cover + long-term memory

WHAT IT ENCODES

HBV PEP for a nonimmune exposed person (needlestick from HBsAg+ source, sexual contact, or neonate of HBsAg+ mother) = HBIG (immediate passive antibody) PLUS the HBV vaccine series (active, durable) given together at different sites, ideally within 24 hours (within 12 hours for newborns).

BOARD TRAP

High-risk HBV exposure in a nonimmune person needs BOTH passive (HBIG now) and active (vaccine for memory) — giving only one is the trap. If the exposed person is already vaccinated and anti-HBs ≥10, no PEP is needed.

8. Passive immunoglobulin shelf

WHAT YOU SEE

Right 'PASSIVE — IMMUNOGLOBULIN' banner: immediate, short, no memory

WHAT IT ENCODES

Passive immunization delivers PREFORMED antibody: IMMEDIATE protection but SHORT-lived (weeks-months) and conferring NO immunologic memory. Used to bridge the gap until active vaccine-induced immunity develops, or in people who can't mount a vaccine response.

BOARD TRAP

Ig = fast but short, with no lasting immunity — it cannot replace vaccination for durable protection. Relying on Ig alone after a high-risk exposure leaves the patient unprotected once the antibody wanes.

9. HBIG — preformed anti-HBs

WHAT YOU SEE

Glowing 'HBIG' jar, 'PREFORMED ANTI-HBs'

WHAT IT ENCODES

HBIG provides immediate, high-titer preformed anti-HBs. Key indications: neonate born to an HBsAg+ mother (HBIG + vaccine within 12 hours of birth) and nonimmune persons after percutaneous/sexual HBV exposure (HBIG + vaccine).

BOARD TRAP

HBIG alone is INCOMPLETE — it must be paired with the vaccine for durable protection, because its preformed antibody fades. Giving HBIG without starting the vaccine series is the wrong answer.

10. HAV immunoglobulin

WHAT YOU SEE

'HAV Ig' jar on passive shelf

WHAT IT ENCODES

HAV immunoglobulin provides immediate passive protection and is the PREFERRED HAV post-exposure prophylaxis for those who shouldn't rely on vaccine: infants <12 months, adults >40 years (where vaccine response is slower/weaker), the immunocompromised, and those with chronic liver disease. Give within 2 weeks of exposure.

BOARD TRAP

Use HAV Ig (not vaccine) for infants <12mo, adults >40, immunocompromised, and chronic liver disease; for healthy people 12mo-40yr the VACCINE is preferred. Mixing up which group gets Ig vs vaccine is the trap (and for >40 or high-risk, both may be given).

11. HEV vaccine limited

WHAT YOU SEE

Small wall placard reading 'HEV: VACCINE EXISTS — LIMITED' tucked above the shelves — the word 'limited' is the bridge: a vaccine technically exists but isn't routinely available.

WHAT IT ENCODES

An HEV vaccine (Hecolin) exists but is licensed/available only in some countries (e.g., China) and is not part of routine US/standard prophylaxis. Prevention elsewhere relies on safe water and avoiding undercooked pork/game.

BOARD TRAP

For US/Israeli boards: assume NO routinely available HEV vaccine. Counsel pregnant travelers to endemic areas on water/food precautions rather than offering a vaccine.

12. Boards rule — risk combo

WHAT YOU SEE

Bottom 'BOARDS RULE' placard: vaccine=slow+long, Ig=fast+short, high risk=both

WHAT IT ENCODES

Master decision rule: VACCINE for lasting protection (slow onset, durable, memory), IMMUNOGLOBULIN for immediate cover (fast onset, short, no memory), and HIGH-RISK exposure gets BOTH together (passive now + active for the future).

BOARD TRAP

The combination (active + passive) is the correct answer for high-risk HBV exposures and for HAV PEP in vulnerable groups — defaulting to only vaccine or only Ig in a high-risk nonimmune exposure is the trap.